ENES 246 · DIGITAL LOGIC DESIGN

Lecture 10

Combinational Analysis and Design



Combinational circuits · the analysis procedure · the design procedure · code converters · seven-segment decoding

Reading: Mano & Ciletti, *Digital Design with an Introduction to the Verilog HDL*, 6th ed. — Chapter 4, §4.1–4.4

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Today’s roadmap

1 §4.1–4.2

Combinational

Outputs depend on inputs only

2 §4.3

Analysis

Circuit → functions → table

3 §4.4

Design

Words → table → circuit

4 §4.4

Code converters

Shared terms; don’t-cares

By the end of class you can…

- Tell a combinational circuit from a sequential one
- Analyze a gate diagram to Boolean functions and a truth table
- Follow the design procedure from a word problem
- Design multiple-output circuits that share gates
- Use unused input codes as don’t-cares

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Agenda for today's session



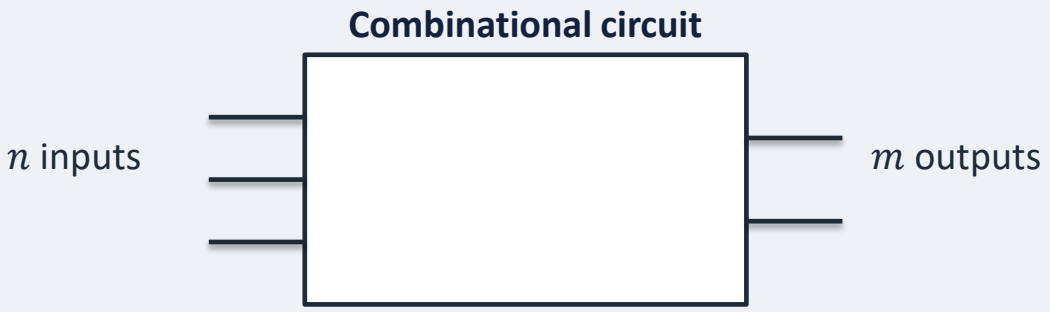
Elapsed time	Topic	In-class work
0:00 – 0:05	Welcome, roadmap	—
0:05 – 0:15	Combinational versus sequential	—
0:15 – 0:40	The analysis procedure	Example 1 · Your turn 1
0:40 – 0:50	Break	
0:50 – 1:10	The design procedure	Example 2
1:10 – 1:40	Code converters and display decoders	Examples 3, 4 · Your turn 2
1:40 – 1:50	Concept check and wrap-up	Exam 1 reminder

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Combinational versus sequential

§4.1–4.2

Combinational



Sequential



Combinational

Outputs depend only on the present inputs.

Described by a truth table or Boolean functions: 2^n rows, m output columns.

Sequential

Outputs also depend on stored state: the circuit has memory and feedback.

Lectures 10–16 are combinational; 18–22 sequential.

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The analysis procedure

§4.3

1. Confirm it is combinational: no feedback, no storage.
2. Label every gate output that is a function of the inputs ($T_1, T_2, \ldots$).
3. Write each label's Boolean function, working from the inputs toward the outputs.
4. Substitute until each output is in terms of the inputs only.
5. Build the truth table, one column per labelled signal; then say what the circuit does.

Two routes

Algebraic: substitute and simplify.

Tabular: evaluate every labelled signal on every row.

The table route is mechanical and hard to get wrong; use the algebra to check it.

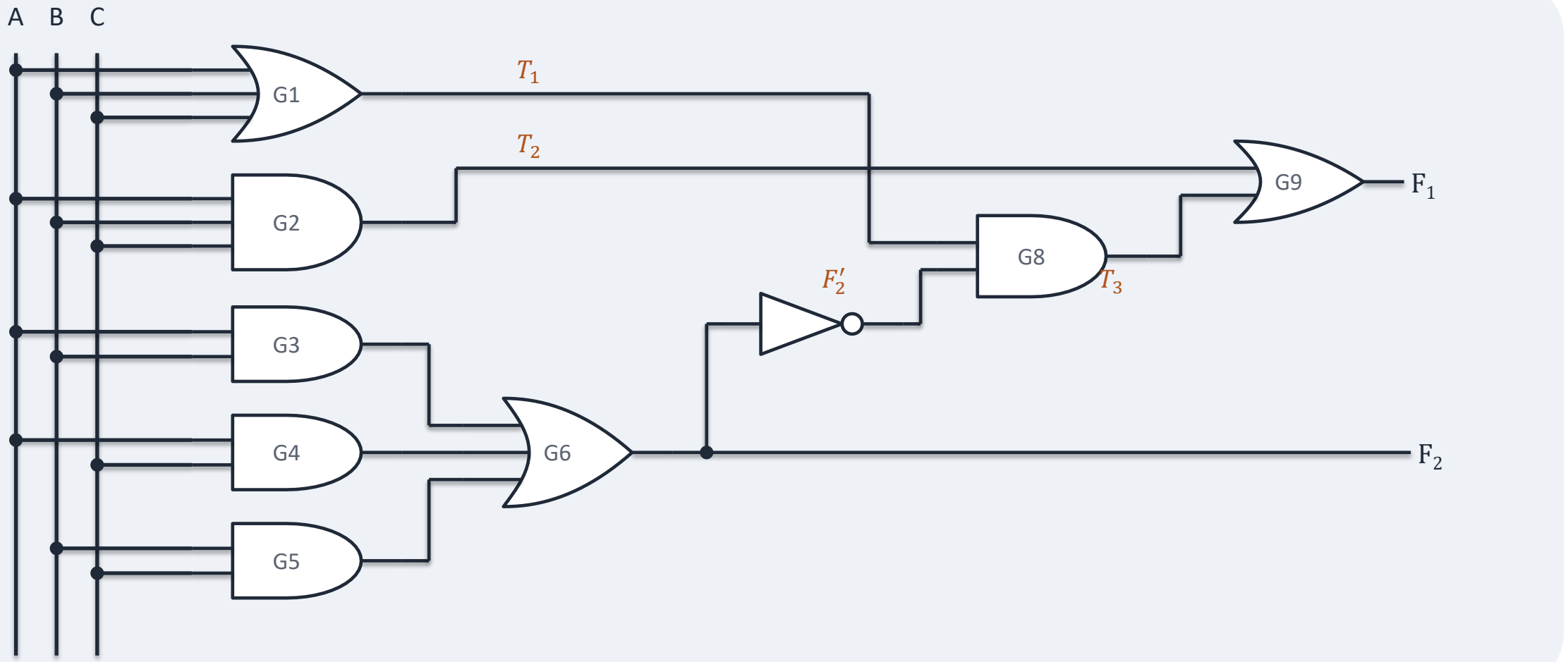
Final step

Recognize the function: an adder, a comparator, a code converter…

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Example 1 — Analyze this circuit

§4.3



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Example 1 — Functions and table

§4.3

A B C	F_2	F_2'	T_1	T_2	T_3	F_1
0 0 0	0	1	0	0	0	0
0 0 1	0	1	1	0	1	1
0 1 0	0	1	1	0	1	1
0 1 1	1	0	1	0	0	0
1 0 0	0	1	1	0	1	1
1 0 1	1	0	1	0	0	0
1 1 0	1	0	1	0	0	0
1 1 1	1	0	1	1	0	1

1. Label each gate output: $T_1 = A + B + C$ (G1), $T_2 = ABC$ (G2)
2. G3–G6: $F_2 = AB + AC + BC$
3. G7 inverts F_2; G8: $T_3 = T_1 F_2'$
4. G9: $F_1 = T_2 + T_3$
5. Fill the table one column at a time, left to right.
6. F_2 is 1 when two or more inputs are 1.
7. F_1 is 1 on 001, 010, 100, 111: an odd number of 1s
8. $F_1 = A \oplus B \oplus C = \Sigma m(1,2,4,7)$
9. **A full adder**: F_1 = sum, F_2 = carry

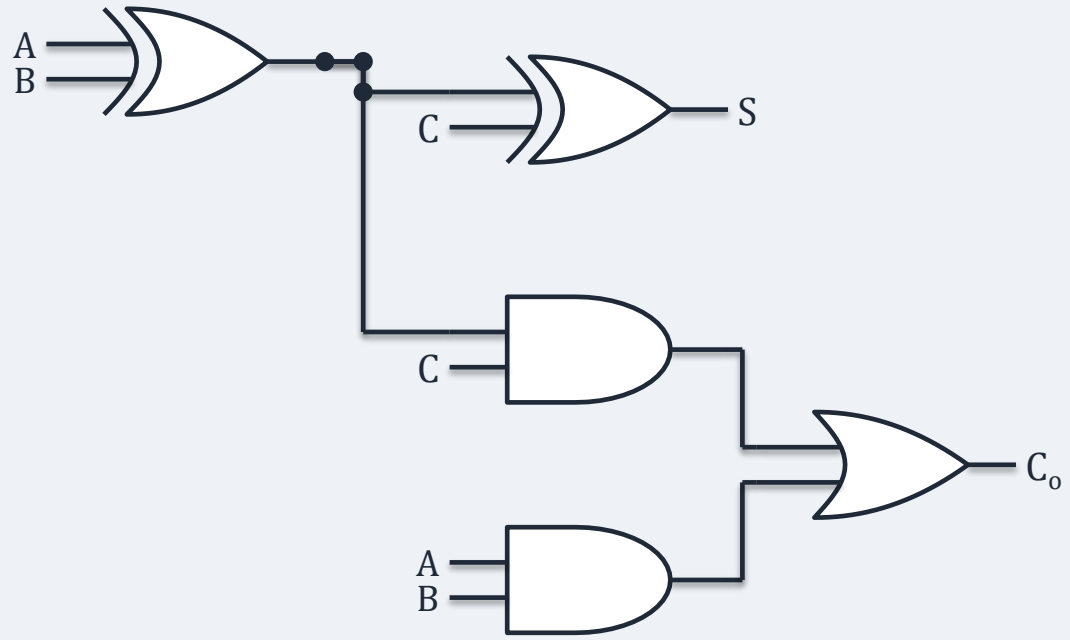
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Your turn 1 — Analyze

§4.3

YOUR TURN

≈ 10 min



(a) Label the internal signals and write S and C_o in terms of A, B, C.

(b) Build the truth table.

(c) Compare it with Example 1. What does this circuit do?

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Your turn 1 — Solution

SOLUTION

A B C	A⊕B	(A⊕B)C	AB	S	C_o
0 0 0	0	0	0	0	0
0 0 1	0	0	0	1	0
0 1 0	1	0	0	1	0
0 1 1	1	1	0	0	1
1 0 0	1	0	0	1	0
1 0 1	1	1	0	0	1
1 1 0	0	0	1	0	1
1 1 1	0	0	1	1	1

1. First XOR: $T = A \oplus B$
2. Second XOR: $S = T \oplus C = A \oplus B \oplus C$
3. The two ANDs: $(A \oplus B)C$ and AB
4. The OR: $C_o = (A \oplus B)C + AB$
5. Fill the table column by column.
6. S matches F_1 and C_o matches F_2 of Example 1
7. **Also a full adder**, with 5 gates instead of 9

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Break

10 minutes — we resume at the 0:50 mark

Next: the design procedure and code converters

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The design procedure

1. From the specification, fix the number of inputs and outputs and name them.
2. Build the truth table that defines each output.
3. Mark input combinations that cannot occur as don’t-cares.
4. Simplify each output (K-map or tabular method).
5. Look for terms the outputs can share.
6. Draw the circuit; verify it against the table.

Constraints

Real designs weigh gates, gate inputs, propagation delay, packages and power.

A minimum for one output is not always the best choice when outputs share terms.

Verify

Evaluate the final circuit on at least two rows, or simulate it.

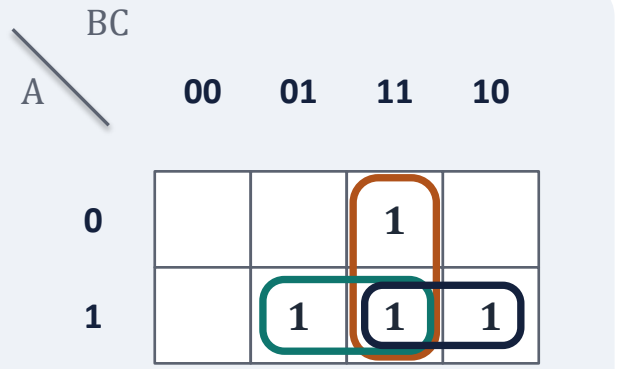
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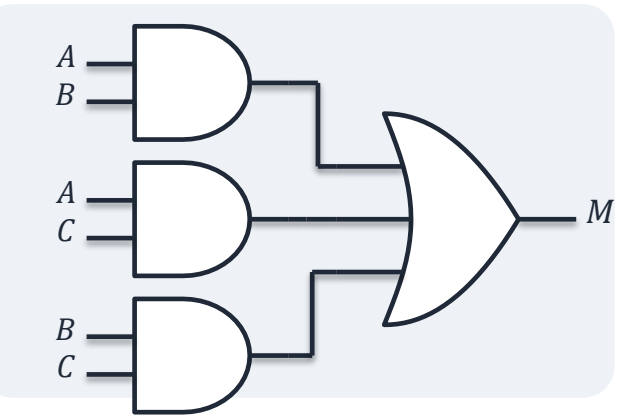
Example 2 — A majority circuit

§4.4

EXAMPLE Three sensors A, B, C. Output $M = 1$ when a majority read 1. Design the minimum circuit.

A B C	M
0 0 0	0
0 0 1	0
0 1 0	0
0 1 1	1
1 0 0	0
1 0 1	1
1 1 0	1
1 1 1	1





1. Inputs A, B, C; one output M
2. $M = 1$ on rows with two or three 1s: 011, 101, 110, 111
3. So $M = \Sigma m(3,5,6,7)$
4. Map: m3, m7 → BC; m5, m7 → AC; m6, m7 → AB
5. Groups may overlap: m7 is used three times.
6. $$M = AB + AC + BC$$
7. NAND-NAND: 3 two-input NANDs + one 3-input NAND

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Example 3 — BCD to excess-3: table

§4.4

Dec	A B C D	w x y z
0	0 0 0 0	0 0 1 1
1	0 0 0 1	0 1 0 0
2	0 0 1 0	0 1 0 1
3	0 0 1 1	0 1 1 0
4	0 1 0 0	0 1 1 1
5	0 1 0 1	1 0 0 0
6	0 1 1 0	1 0 0 1
7	0 1 1 1	1 0 1 0
8	1 0 0 0	1 0 1 1
9	1 0 0 1	1 1 0 0

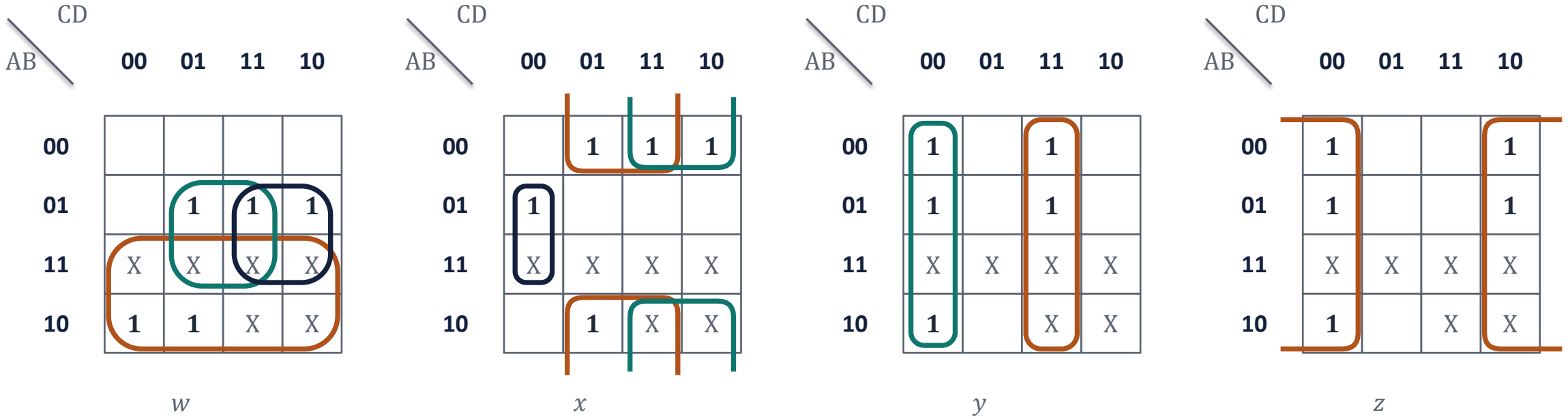
Inputs 1010–1111: don’t-care outputs

1. Input: one BCD digit $ABCD$ (A = MSB)
2. Output: excess-3 code $wxyz$ = digit + 3
3. Only 0000–1001 occur: $d(10,11,12,13,14,15)$ for every output
4. Check: $5 + 3 = 8 \rightarrow 1000$; $9 + 3 = 12 \rightarrow 1100$ ✓
5. Next: one K-map per output

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Example 3 — Four maps

§4.4



$$w = A + BC + BD \quad x = B'C + B'D + BC'D' \quad y = CD + C'D' \quad z = D'$$

Each is a minimum on its own map.

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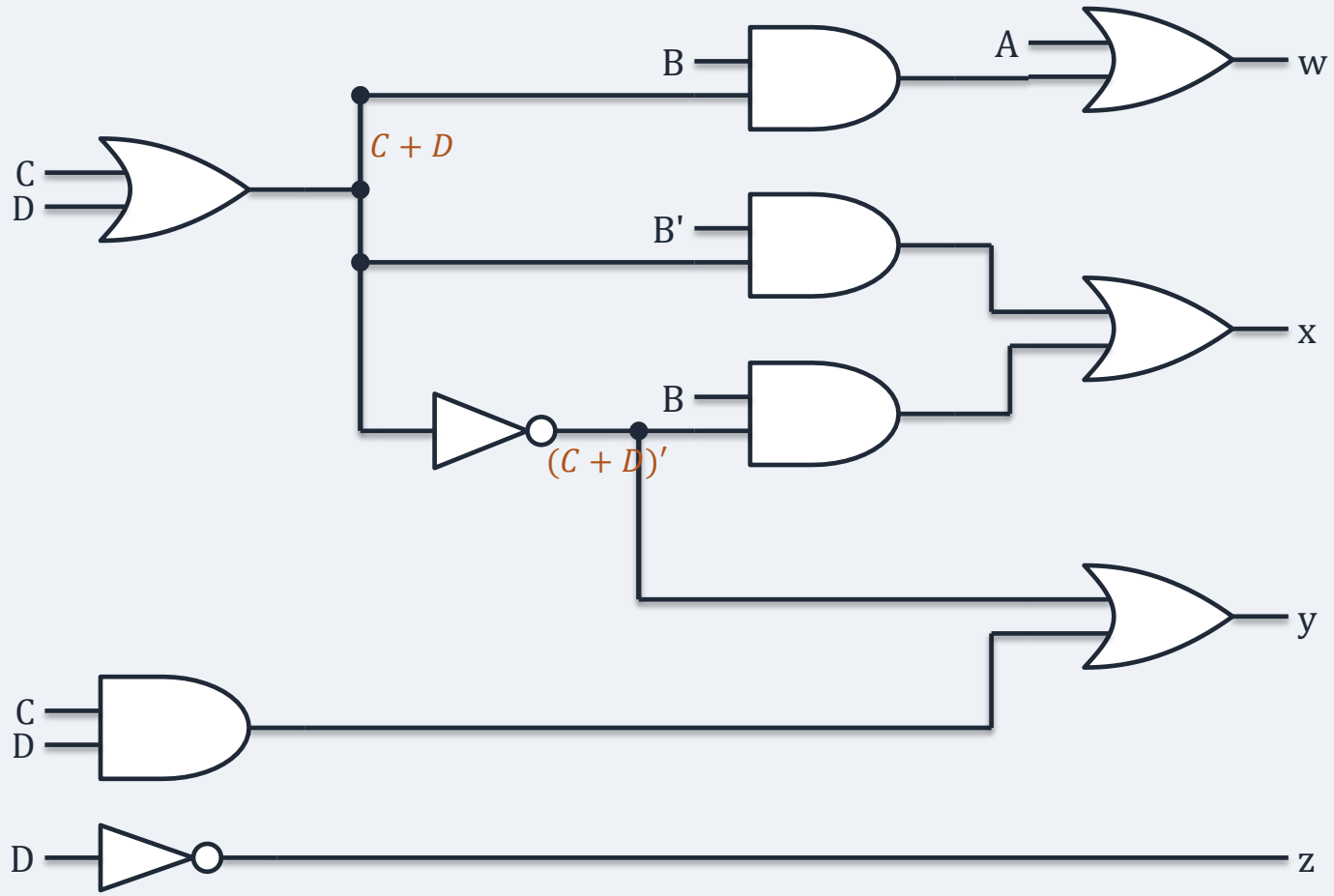
Example 3 — Reading the maps

§4.4

1. w: octet of rows 11, 10 (X as 1): $A = 1 \rightarrow A$
2. w: quad m5, m7, m13, m15: $B = 1, D = 1 \rightarrow BD$
3. w: quad m6, m7, m14, m15: $B = 1, C = 1 \rightarrow BC$
4. So $w = A + BC + BD$
5. x: quad m1, m3, m9, m11: $B = 0, D = 1 \rightarrow B'D$
6. x: quad m2, m3, m10, m11: $B = 0, C = 1 \rightarrow B'C$
7. x: pair m4, m12: $B = 1, C = 0, D = 0 \rightarrow BC'D'$
8. So $x = B'C + B'D + BC'D'$
9. y: column 11 $\rightarrow CD$; column 00 $\rightarrow C'D'$
10. So $y = CD + C'D'$
11. z: the octet of columns 00 and 10: $D = 0$
12. So $z = D'$

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Example 3 — Sharing $C + D$



Factor for sharing

1. $w = A + BC + BD = A + B(C + D)$
2. $x = B'C + B'D + BC'D'$

   $= B'(C + D) + BC'D'$
3. DeMorgan: $C'D' = (C + D)'$

   $x = B'(C + D) + B(C + D)'$
4. $y = CD + C'D' = CD + (C + D)'$
5. $z = D'$

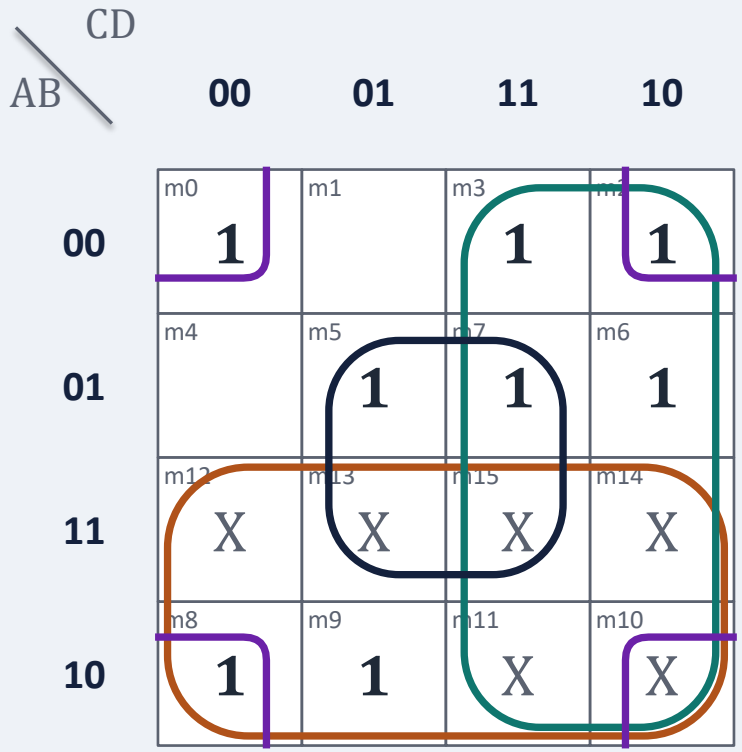
One $C + D$ gate and its complement feed three outputs.

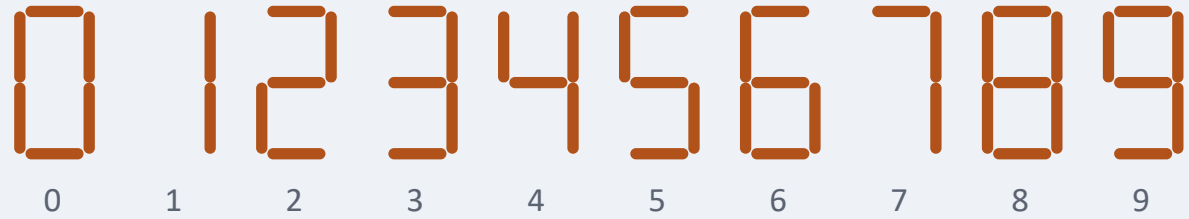
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Example 4 — Seven-segment segment a

§4.4

EXAMPLE A BCD digit $ABCD$ drives a seven-segment display. Design the output for segment a (active high).





1. From the displays: a is off only for 1 and 4
2. $a = \Sigma m(0,2,3,5,6,7,8,9) + d(10, \ldots, 15)$
3. Octet of rows 11, 10 (X as 1) → A
4. Octet of columns 11, 10 → C
5. Quad m5, m7, m13, m15 → BD; four corners → $B'D'$
6. $$a = A + C + BD + B'D'$$

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Your turn 2 — Binary to Gray

§4.4

YOUR TURN

≈ 10 min

Design a circuit that converts a 3-bit binary number $B_2B_1B_0$ to its Gray code $G_2G_1G_0$.

(a) Build the truth table.

(b) Use a K-map for G_1 and for G_0.

(c) Draw the circuit with the fewest gates.

Hint: look for a checkerboard.

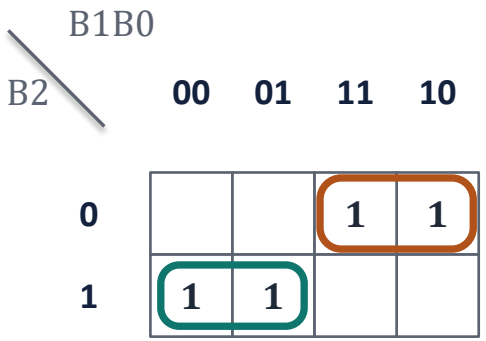
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Your turn 2 — Solution

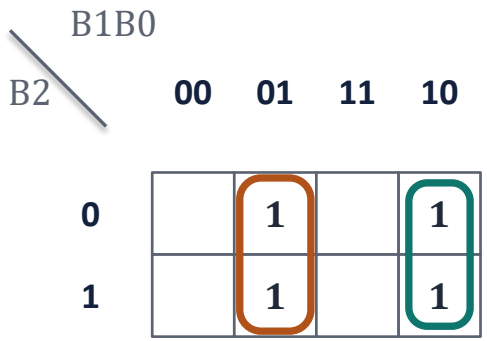
§4.4

SOLUTION

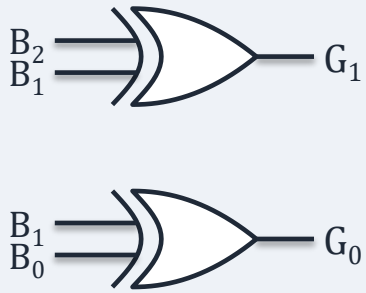
$B_2B_1B_0$	$G_2G_1G_0$
000	000
001	001
010	011
011	010
100	110
101	111
110	101
111	100



G_1



G_0



$G_2 = B_2$: a wire

1. Table: $G_2 = B_2$, $G_1 = B_2 \oplus B_1$, $G_0 = B_1 \oplus B_0$
2. G_2 column equals B_2: a wire
3. G_1 map: 1s at 2, 3, 4, 5
4. m2, m3 → $B_2'B_1$; m4, m5 → B_2B_1'
5. $G_1 = B_2'B_1 + B_2B_1' = B_2 \oplus B_1$
6. G_0: m1, m5 → $B_1'B_0$; m2, m6 → B_1B_0'
7. $G_0 = B_1'B_0 + B_1B_0' = B_1 \oplus B_0$
8. **Two XOR gates in all**

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Concept check

1. How can you tell from a diagram that a circuit is combinational?

2. Why label every internal gate output during analysis?

3. Why are inputs 1010–1111 don’t-cares in a BCD converter?

4. When can a non-minimal form for one output give a cheaper circuit?

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Key takeaways

Combinational

outputs = f(inputs)

Analysis

label → substitute → table

Design

table → maps → share → draw

Majority

$$AB + AC + BC$$

Excess-3

$$z = D', y = CD + C'D'$$

Segment a

$$A + C + BD + B'D'$$

Analysis and design are the same table, read in opposite directions.

Homework #6 due at the start of Lecture 13

This afternoon's lab: Lab D — XOR gates, simplification and parity

Next: Exam 1, then decoders and encoders

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